

PHYSICS CH.12 — UNITS AND MEASUREMENTS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

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1. Fundamental & Derived SI Units

- SI system mein 7 base/fundamental units hain: meter (length), kilogram (mass), second (time), ampere (current), kelvin (temperature), mole (amount), candela (luminous intensity). [Q1495, Q1544]
- Newton (N) = derived unit; $1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$. [Q1495]
- Joule (J) = derived unit; $1 \text{ J} = 1 \text{ N} \cdot \text{m}$. [Q1495, Q1551]
- Watt (W) = derived unit; $1 \text{ W} = 1 \text{ J/s}$. [Q1495, Q1552]
- Pascal (Pa) = derived unit; $1 \text{ Pa} = 1 \text{ N/m}^2$. [Q1495, Q1556]
- Coulomb (C) = derived unit; $1 \text{ C} = 1 \text{ A} \cdot \text{s}$. [Q1522, Q1525]
- Volt (V) = derived unit; $1 \text{ V} = 1 \text{ J/C}$. [Q1531, Q1540]
- Ohm (Ω) = derived unit; $1 \Omega = 1 \text{ V/A}$. [Q1526, Q1528]
- Hertz (Hz) = derived unit; $1 \text{ Hz} = 1 \text{ cycle/s}$. [Q1504, Q1546]
- Farad (F) = capacitance ka unit. [Q1534]
- Henry (H) = inductance ka unit. [Q1535]
- Weber (Wb) = magnetic flux ka unit. [Q1538]
- Tesla (T) = magnetic field ka unit; $1 \text{ T} = 1 \text{ Wb/m}^2$. [Q1539, Q1541]
- Lumen (lm) = luminous flux ka unit. [Q1544]
- Lux (lx) = illuminance ka unit; $1 \text{ lx} = 1 \text{ lm/m}^2$. [Q1544]
- Area ka unit m^2 ; volume ka m^3 . [Q1496]

2. Electrical Units, Energy & Power

- 1 kWh (commercial unit) = $3.6 \times 10^6 \text{ J} = 1 \text{ kW} \times 1 \text{ hour}$. [Q1490, Q1532, Q1545]
- Electric power ke valid formulas: $P = VI = I^2R = V^2/R$; unit Watt (W). [Q1532, Q1537, Q1548]

- Electrical energy: $E = Pt = VIt$; commercial unit kWh. [Q1532]
- 60 W bulb 1 hour $\rightarrow$ Energy = $60 \times 3600 = 216000 \text{ J} = 216 \text{ kJ}$. [Q1497]
- 100 W bulb 10 hours $\rightarrow$ 1 kWh. [Q1492, Q1545]
- 200 W bulb 5 hours $\rightarrow$ 1 kWh. [Q1545]
- 2000 W heater 30 min $\rightarrow$ 1 kWh. [Q1514]
- 100 W bulb 24 hours $\rightarrow$ 2.4 kWh. [Q1524]
- 500 W appliance 2 hours $\rightarrow$ 1 kWh. [Q1548]
- Current ka SI unit Ampere (A) = C/s. [Q1525, Q1527]
- 1 A current = 1 C charge per second. [Q1527]
- Resistance ka SI unit Ohm (Ω). [Q1526, Q1528]
- Ohm's Law: $V = IR$; $R = V/I$. [Q1528]
- Resistivity ka unit Ohm-meter ($\Omega\cdot\text{m}$). [Q1529]
- Conductance ka unit Siemens (S) = $1/\Omega$. [Q1533]
- Electric field ka unit V/m ya N/C. [Q1536]
- EMF (Electromotive force) ka unit bhi Volt (V) hota hai. [Q1531]
- Radiation measure karne ke liye Geiger-Muller counter use hota hai — Voltmeter/Ammeter/Spectrometer nahi. [Q1523]

3. Work, Power & Energy Conversions

- Work/Energy ka SI unit Joule (J). [Q1493, Q1498, Q1501, Q1551]
- Power ka SI unit Watt (W) = J/s. [Q1499, Q1500, Q1552]
- 1 MW = 10^6 W ; 1 GW = 10^9 W . [Q1505]
- 1 Horsepower (HP) = $746 \text{ W} = 0.746 \text{ kW}$. [Q1494, Q1502, Q1503, Q1506]
- 1.5 HP = 1119 W ; 2 HP = 1492 W ; 5.5 HP = 4103 W . [Q1502, Q1503, Q1506]
- 1 erg = 10^{-7} J (CGS unit of work). [Q1553]
- 1 dyne = 10^{-5} N (CGS unit of force). [Q1554]
- 1 eV (electron volt) = $1.6 \times 10^{-19} \text{ J}$; 1 MeV = $10^6 \text{ eV} = 1.6 \times 10^{-13} \text{ J}$. [Q1555]
- 1 calorie = $4.186 \text{ J} \approx 4.2 \text{ J}$. [Q1521]

4. Mechanical Units (Force, Pressure, Density, Speed)

- Force ka SI unit Newton (N). [Q1495, Q1511, Q1512]
- 1 kgf (kilogram force) = $9.8 \text{ N} \approx 10 \text{ N}$. [Q1512]
- Momentum ka unit $\text{kg}\cdot\text{m/s}$ ya $\text{N}\cdot\text{s}$. [Q1493]
- Acceleration ka unit m/s^2 . [Q1501]
- Speed/velocity ka SI unit m/s ; distance ka meter (m). [Q1508, Q1509, Q1510]
- 1 km/h = $5/18 \text{ m/s}$. [Q1508]
- Speedometer instantaneous speed measure karta hai. [Q1510]
- Odometer total distance measure karta hai. [Q1509]
- Pressure ka SI unit Pascal (Pa) = N/m^2 . [Q1495, Q1513, Q1556, Q1560]
- 1 bar = 10^5 Pa ; 1 atm = $1.013 \times 10^5 \text{ Pa} = 101.325 \text{ kPa}$. [Q1556]

- 1 torr = 133.322 Pa; 1 mm Hg $\approx$ 133 Pa. [Q1556]
- Density ka unit kg/m^3 ; CGS mein g/cm^3 . [Q1515, Q1557]
- $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$. [Q1557]
- Relative density = density of substance / density of water; no unit — momentum, density, pressure sab ka unit hota hai lekin relative density dimensionless hai. [Q1515, Q1530]
- Specific volume = $1/\rho$; unit m^3/kg . [Q1558]
- Surface tension ka unit N/m ya J/m^2 . [Q1559]
- Torque/Moment of force ka unit N·m; lekin ye work se alag hai — direction alag hoti hai. [Q1551]
- 1 Poise = $0.1 \text{ Pa}\cdot\text{s}$ (CGS viscosity unit). [Q1561]
- 1 Stokes = $10^{-4} \text{ m}^2/\text{s}$ (CGS kinematic viscosity). [Q1562]

5. Heat & Temperature Units

- Temperature ka SI unit Kelvin (K) — base unit. [Q1517, Q1520]
- $0^\circ\text{C} = 273.15 \text{ K}$; $100^\circ\text{C} = 373.15 \text{ K}$. [Q1520]
- Specific heat capacity ka unit $\text{J}/(\text{kg}\cdot\text{K})$ ya $\text{J}/(\text{kg}\cdot^\circ\text{C})$. [Q1518]
- Latent heat ka unit J/kg . [Q1519]

6. Magnetic & Light Units

- Magnetic flux ka unit Weber (Wb). [Q1538]
- Magnetic field (B) ka unit Tesla (T) = Wb/m^2 . [Q1539, Q1541]
- 1 Gauss = 10^{-4} Tesla. [Q1539]
- Magnetic field intensity ka unit A/m. [Q1542]
- Magnetic permeability ka unit H/m ya $\text{T}\cdot\text{m/A}$. [Q1543]
- Luminous intensity ka SI unit Candela (cd) — base unit. [Q1544, Q1549]
- Luminous flux ka unit Lumen (lm). [Q1544]
- Illuminance ka unit Lux (lx) = lm/m^2 . [Q1544]

7. Sound & Wave Units

- Frequency ka unit Hertz (Hz). [Q1504, Q1546]
- Time period ka unit second (s). [Q1507]
- $1 \text{ MHz} = 10^6 \text{ Hz}$; $1 \text{ GHz} = 10^9 \text{ Hz}$. [Q1504]
- Wavelength ka unit meter (m); commonly nanometer (nm). [Q1547]
- $1 \text{ nm} = 10^{-9} \text{ m}$. [Q1547]
- Audible range: 20 Hz se 20,000 Hz (20 kHz). [Q1504]
- Ultrasound: $>20 \text{ kHz}$; Infrasound: $<20 \text{ Hz}$. [Q1504]

8. Lens & Optical Units

- Magnification (m) ka koi unit nahi hai — ye dimensionless/ratio hota hai ($m = v/u = h_i/h_o$). [Q1491, Q1550]
- Dioptre (D) lens power ($P = 1/f$) ka unit hai; magnification ka nahi. [Q1491, Q1516]
- $f = 50 \text{ cm} = 0.5 \text{ m} \rightarrow P = +2 \text{ D}$ (convex). [Q1516]
- $f = -40 \text{ cm} = -0.4 \text{ m} \rightarrow P = -2.5 \text{ D}$ (concave). [Q1516]

9. ⚠ Traps / Key Differences Summary (MCQ Q1490–Q1562)

- ⚠ Trap: Magnification ka unit nahi hota — ye dimensionless ratio hai. [Q1491, Q1550]
- ⚠ Trap: kWh energy ki unit hai, power ki nahi — confuse mat hona. [Q1490]
- ⚠ Trap: 1 HP = 746 W yaad rakhna — 1000 W nahi. [Q1494]
- ⚠ Trap: Newton derived unit hai — fundamental nahi. [Q1495]
- ⚠ Trap: Relative density ka unit nahi hota. [Q1515, Q1530]
- ⚠ Trap: 1 calorie = 4.2 J approx — exact 4.186 J hai. [Q1521]
- ⚠ Trap: $0^\circ\text{C} = 273 \text{ K}$ approx, exact 273.15 K hai. [Q1520]
- ⚠ Trap: EMF aur Voltage dono ka unit Volt (V) hota hai — same unit, different concept. [Q1531]
- ⚠ Trap: Torque ka unit N·m hai lekin ye Joule nahi hota; torque vector hai, work scalar. [Q1551]
- ⚠ Trap: 1 Gauss = 10^{-4} Tesla — confuse mat hona. [Q1539]
- ⚠ Trap: Candela (cd) base unit hai — lumen derived unit hai. [Q1544]
- ⚠ Trap: $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$ — water ka density $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$. [Q1557]
- ⚠ Trap: $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ — atomic level energy ke liye use hota hai. [Q1555]
- ⚠ Trap: $1 \text{ bar} = 10^5 \text{ Pa} \approx 1 \text{ atm}$ — standard atmospheric pressure. [Q1556]
- ⚠ Trap: Radiation measure karne ka device Geiger-Muller counter hai, Voltmeter/Ammeter/Spectrometer nahi. [Q1523]

10. Numerical: Acceleration & Basic Mechanical Units

- Acceleration ka SI unit m/s^2 hota hai; ye velocity ke rate of change ko batata hai — formula $a = (v - u)/t$. [Q1563]
- Acceleration due to gravity $g = 9.8 \text{ m/s}^2$ hota hai. [Q1563]

11. Numerical: Temperature Scale Conversions

- Kelvin se Celsius mein convert karne ke liye 273 subtract karte hain: $C = K - 273$. [Q1564]
- Teenon scales ka relation: $(F - 32)/180 = C/100 = (K - 273)/100$. [Q1564]

12. Numerical: Power & Energy Unit Conversions

- $1 \text{ kW} = 1000 \text{ W}$; isliye $2500 \text{ W} = 2.5 \text{ kW}$. [Q1565]
- $1 \text{ Watt} = 1 \text{ Joule/second}$ (1 J/s); power ko energy per unit time define karte hain. [Q1569]
- $1 \text{ kWh} = 3.6 \times 10^6 \text{ J} = 3,600,000 \text{ J}$; calculation: $1000 \text{ W} \times 3600 \text{ s}$. [Q1566, Q1567]
- $4.2 \text{ kWh} = 15.12 \times 10^6 \text{ J}$; multiply karna padta hai: $4.2 \times 3.6 \times 10^6 \text{ J}$. [Q1568]
- $1 \text{ Joule} = 10^7 \text{ erg}$; CGS unit of work. [Q1569]

13. Numerical: Circular Motion — Velocity Formula

- Uniform circular motion mein velocity formula: $v = 2\pi r/t$; jahan r radius aur t ek complete round ka time hai. [Q1570]
- Circumference $2\pi r$ ko time t se divide karke velocity milti hai. [Q1570]

14. Numerical: Formula Cheat Sheet

- Acceleration unit: m/s^2 .
- $K \rightarrow ^\circ C$: $C = K - 273$.
- $1\text{ kW} = 1000\text{ W}$.
- $1\text{ W} = 1\text{ J/s}$.
- $1\text{ kWh} = 3.6 \times 10^6\text{ J}$.
- $1\text{ J} = 10^7\text{ erg}$.
- Circular velocity: $v = 2\pi r/t$.

15. ⚠ Traps / Key Differences Summary (Numerical Q1563–Q1570)

- ⚠ Trap: Kelvin se Celsius mein convert karte waqt 273 ghatana hota hai, jodna nahi. [Q1564]
- ⚠ Trap: $1\text{ kWh} = 3.6 \times 10^6\text{ J}$ yaad rakhna — 3600 J ya 360000 J options confuse karne ke liye hote hain. [Q1566, Q1567]
- ⚠ Trap: 4.2 kWh ko Joule mein badalne ke liye $4.2 \times 3.6 \times 10^6$ karna hota hai, sirf 3.6 se multiply nahi. [Q1568]
- ⚠ Trap: $1\text{ W} = 1\text{ J/s}$ hai — erg/s nahi; $1\text{ J} = 10^7\text{ erg}$ hota hai. [Q1569]
- ⚠ Trap: Circular motion mein velocity $v = 2\pi r/t$ hai — $\pi r/t$ ya $2\pi r^2/t$ galat hain. [Q1570]
- ⚠ Trap: Acceleration ka unit m/s^2 hai — m/s (velocity) ya ft/s galat options hain. [Q1563]